

The empirical recurrence for OEIS A254265, recovered from its tail

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Abstract

OEIS A254265 records “Empirical recurrence of order 53” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 388$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 15$, so the recurrence is proved for every $n \geq 68$. Its last coefficient is $c_{53} = 39582418599936 \neq 0$, so no further tail satisfies anything shorter; any order-53 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254265 is “Number of $(6+2)X(n+2)$ 0..1 arrays with every $3X3$ subblock sum of the two maximums of the diagonal and antidiagonal minus the sum of the minimums of the central row and column nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

1021428, 3647288, 6729896, 29231312, 87535036, 325742026, 1249946456, 4118577869, ...

The entry states:

Empirical recurrence of order 53 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:35:57, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 388$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 388$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 53: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 15$ is the first whose minimal order is exactly the stated 53.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 776$ exact terms from $n = 15$ on, it returns order 53 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{53} c_i a(n-i),$$

c_1	9	c_{15}	73768950	c_{29}	−50976973360128	c_{43}	893218107424768
c_2	15	c_{16}	−1198501300	c_{30}	−118261147783168	c_{44}	701947577368576
c_3	−244	c_{17}	126111570	c_{31}	14178833739776	c_{45}	−91463207616512
c_4	−519	c_{18}	11971716564	c_{32}	262631452606464	c_{46}	−484522776854528
c_5	3757	c_{19}	3135107456	c_{33}	202067198590976	c_{47}	−444576359776256
c_6	18049	c_{20}	−96646824480	c_{34}	−261918263967744	c_{48}	−9388798509056
c_7	−46658	c_{21}	−115929271328	c_{35}	−531450439041024	c_{49}	361241109331968
c_8	−299107	c_{22}	515862678752	c_{36}	−157109323366400	c_{50}	175234665676800
c_9	246941	c_{23}	1225759142816	c_{37}	527660045828096	c_{51}	−141836999983104
c_{10}	3278949	c_{24}	−1441157482368	c_{38}	746651995602944	c_{52}	−79164837199872
c_{11}	1225632	c_{25}	−7129556940544	c_{39}	183661889585152	c_{53}	39582418599936
c_{12}	−26649409	c_{26}	−1184446680576	c_{40}	−745302751444992		
c_{13}	−20091781	c_{27}	25575998852736	c_{41}	−980865211432960		
c_{14}	163830871	c_{28}	27925710302464	c_{42}	−56528027516928		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 68$, and it is the recurrence OEIS A254265 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 15$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{53} - c_1 x^{53-1} - \dots - c_{53}$. Its constant term is $-c_{53} = -39582418599936$, which is not zero, so

$q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 53 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 53, so $\deg q = 53$, and it is monic. It holds from some index t on. From $\max(t, 15)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 53, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 53, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 39582418599936.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-53 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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